Brian Wonjun Lee

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Education

University of Pennsylvania Philadelphia, PA
MSE in Computer Graphics & Game Technology (CGGT), Submatriculation *Expected May 2028*
BA in Computer Science & Design (Double Major), Benjamin Franklin Scholars *Expected May 2027*
Relevant Coursework: Interactive Computer Graphics, Computer Animation, 3-D Computer Modeling, Introduction to Algorithms, Introduction to Computer Systems, Programming Languages & Techniques

Experience

Software Engineer Intern, Aleph Lab (Y Combinator) *June 2026 – Present · San Francisco (Remote)*

- Built and **shipped to production** a recency-segmented retention system (Supabase triggers, Deno edge functions, FCM) at a 4-engineer YC startup, with a treatment/holdout A/B that drove the product's **first measured win-back → paid conversion**
- Caught a win-back targeting flaw by validating against production data — repointing from child-role to cross-account activity for a **~2.7× gain in reachable users**
- Packaged a production AI agent as an **installable service** behind a typed SDK — 5 versioned wire contracts, **136 tests**; a fresh AI agent integrated a new environment in **0 fix-loops**
- **Root-caused a production auth outage** from a JWT signing-key migration and shipped the fix, restoring the guarded notification pipeline (401 → 200)

Software Engineer Intern, BitMango (Puzzle1 Studio) *June 2025 – Aug 2025 · Pangyo, South Korea*

- Maintained a live portfolio of **50+ Unity mobile game titles**, improving UI/UX and core gameplay stability across deployments
- Resolved **100+ QA-reported bugs** by modifying Unity prefabs and C# scripts, shipping fixes in production releases
- Upgraded SDKs and platform modules **fleet-wide** to meet evolving app-store requirements and sustain production availability

Project Lead & Product Designer, Penn Spark *Jan 2025 – Present · Philadelphia, PA*

- Led a **15+-person cross-functional team** of designers and developers to ship client-facing **0-to-1 products**
- Defined **end-to-end user flows**, interaction patterns, and reusable interface systems across multiple deliverables
- Shipped polished, **user-centered features** across semester-long client project cycles

Military Interpreter, Capital Defense Command (ROK Army) *Dec 2022 – June 2024 · Seoul, South Korea*

- Provided **real-time English-Korean interpretation** in high-security, multi-agency environments for international stakeholders
- Translated **sensitive documents and briefings** with precision, enabling coordination across government and defense teams

Database Engineer Intern, IT-Farm *June 2022 – Aug 2022 · Seongnam, South Korea*

- Designed and implemented a SQL database prototype managing **1M+ data entities** using Oracle and SQLite
- Processed and organized semiconductor datasets from **20+ client companies**, including Samsung, SK, and LG
- Wrote and **optimized SQL queries** to improve retrieval and management efficiency across large-scale datasets

Projects

CareerOS | *TypeScript, Node.js, LLM/Agent tooling, Astro* *2026*

- Built an **AI-assisted career platform**: automated role sourcing across **~110 ATS boards** surfacing 5,000+ live roles
- Engineered deterministic fit-ranking and a **cross-device application tracker** with live Google Sheet sync
- Shipped an **agentic morning-brief** and answer-drafting assistant — a personal career operating system

Capsule | *React, Three.js (R3F), MongoDB, AWS S3* *Jan 2025 – Apr 2025*

- Built a **full-stack web app** (team of 4) for creating interactive 3D time capsules with media storage and timed unlocking
- Engineered immersive 3D scenes with **React Three Fiber** for real-time rendering, animation, and direct-manipulation interaction
- Designed the backend with **MongoDB** and scalable media storage on **AWS S3**, wiring authentication and timed-release logic

Mini Minecraft | *C++, OpenGL, GLSL* *Apr 2026*

- Engineered a **voxel game engine from scratch** in C++/OpenGL — procedural world generation across **7 biomes** via layered Perlin/FBM/Voronoi noise, plus 3D cave systems and fluid simulation
- Designed **chunk-based streaming with frustum culling** and a data-oriented mesh layout for real-time performance at scale

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, C++, C#, SQL, HTML/CSS, OCaml
Frameworks & Libraries: React, Next.js, Node.js, Astro, Tailwind CSS, Three.js, React Three Fiber, JUnit
Developer Tools: Git, GitLab, AWS, MongoDB, SQLite, Oracle, Figma, VS Code, IntelliJ
Graphics & 3D: OpenGL, GLSL, Unreal Engine 5, Unity, Maya, Blender